HEAPLINE · TRACEABILITY MATRIX

HeapLine traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heapline		
Source	docs/03b-traceability.md @ 0055524		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-11-03	Amish Chadha	Generated by .kit/trace.py
0.5	2026-11-03	Amish Chadha	Generated by .kit/trace.py
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0.1	2026-11-03	Amish Chadha	Generated by .kit/trace.py

HeapLine traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 15 user requirements, 15 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Deploy fast	HPL-CAL-001, HPL-BLD-001, HPL-STM-001	Timed drills with the partner cooperative	Met on paper
UR-002	PR-002	Probe depth	HPL-CAL-001, HPL-BLD-001	Probe trials on a waste test heap, including whether the sleeve friction rises with the 18 mm tip's smaller clearance	Met on paper
UR-003	PR-003	Crawl board support	HPL-CAL-001, HPL-BLD-001	Load tests on representative fill	Met on paper
UR-004	PR-004	Capstan pull	HPL-CAL-001, HPL-BLD-001	Pull test on CalRig	Met on paper
UR-005	PR-005	Casualty transport	HPL-CAL-001, HPL-BLD-001	Drill with manikin	Met on paper under the drill-card rule
UR-006, UR-012	PR-006	Stop rule understood	HPL-CAL-001, HPL-BLD-001	Post-drill check	To verify at TRL 4
UR-007	PR-007	Storage life	HPL-CAL-001, HPL-BLD-001, HPL-STM-001	Annual inspection log at the pilot site	Met by design
UR-008	PR-008	Portable	HPL-CAL-001, HPL-BLD-001, HPL-STM-001	Weighing; timed carry in two waves	Met on paper in two waves
UR-009	PR-009	Cost	HPL-CAL-001	Costed bill of materials	Met on paper as restated
UR-010	PR-010	Overload limit	HPL-CAL-001, HPL-BLD-001, HPL-OPT-001	Shear pin release tests on a batch (TRL 4)	Met on paper
UR-010	PR-011	Holds when let go	HPL-CAL-001, HPL-BLD-001, HPL-STM-001, HPL-OPT-001	Release test at working pull (TRL 4)	Met on paper
UR-014	PR-012	Light without power	HPL-CAL-001	Inspection at the monthly check (BOM line 25)	Met by design
UR-011	PR-013	Repair with common stock	HPL-CAL-001	Review of the BOM and HPL-BLD-001	Met by design
UR-013	PR-014	Not certified equipment	HPL-CAL-001	Document review	Met by design
UR-015	PR-015	Shared capstan arrival	HPL-CAL-001, HPL-STM-001	Timed capstan wave drill at TRL 4	Met on paper

Gaps

None.